Procedure, Not Property: SAE Features Whose Ablation Improves Multilingual Reasoning Are Not (Just) Language Features

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MIT 6.8610: Quantitative Methods for Natural Language Processing, Spring 2026

1. The Question

Counter-intuitively, removing language-specific information from a model's hidden state can *improve* multilingual reasoning. We use sparse autoencoders (SAEs) [1], which decompose hidden states into interpretable individual features, to ask what those removed representations actually are.

Zhao et al. (NeurIPS 2025)

Zhao et al. [2] showed that multilingual reasoning is partly degraded by *language-specific representations interfering with reasoning*. Their method projects out a per-language mean subspace (rank-3 SVD) at inference: $\hat{h} = h - \lambda M_s M_s^\top h$. Across 10 models and 10 languages: consistent gains up to +12.8pp, especially for low-resource languages.

Our extension

Zhao's method operates on entire subspaces and cannot identify individual features or explain the mechanism. We use SAEs to decompose the effect into individual interpretable features and characterize both the **mechanism** and the **nature** of the ablated representations.

The question

What are these features doing, and what are they encoding?

Using SAEs on Gemma 3 4B IT, we identify candidate features via monolinguality (ν) intersected with a supervised language probe. The "language" label is induced by this selection criterion; no human inspects a feature and assigns it. Ablation improves reasoning for en (+13.2pp) and es (+6.8pp). Three lines of evidence characterize this induced label as a **procedural tag** whose scope is determined by the selection method that produced it.

2. SAE Language Feature Extraction

We pass 250 MGSM math problems [3] $\times$ 5 languages through Gemma 3 4B IT, extract last-token residual-stream activations, and encode them through Gemma Scope 2 SAEs [4] (16k-width, layers {9, 17, 22, 29}).

Two identification methods

- **Method A. Monolinguality (ν)** [5]: $\nu_s^L = \mu_s^L - \gamma_s^L$ (mean activation on target language minus mean across other languages). Top-50 per language.
- **Method B. Supervised probe:** Logistic regression on SAE feature vectors (5-way language classification, accuracy ceiling $\sim$0.88). Top-50 per language by coefficient magnitude.

Candidates $=$ A $\cap$ B intersection. Methods mostly disagree (Jaccard 0.03–0.33); the intersection selects features that *both* methods agree are language-differential.

Identifying reasoning features

MGSM contains the same 250 math problems translated into each language. A feature is tagged as **reasoning** if it activates (above threshold) in *all 5 languages for the same problem* on $\geq$10% of problems. This criterion selects features whose activation tracks mathematical content across language translations. Of 16,384 SAE features at layer 17, 19 satisfy it (0.12%).

3. Ablation Results

A. Downstream accuracy

Ablate top-k candidate features per language at layer 17 using directional (QR-orthogonalized) projection. Prior work shows single-feature SAE steering already controls model output language [6]; we extend this to multi-feature ablation and ask whether *reasoning* also responds. k=20 selected on 50-prompt dev split.

MGSM accuracy after language-feature ablation (layer 17, k=20)

Lang	Baseline	Zhao SVD	SAE	Δ vs base	Δ vs Zhao
en	0.580	0.632	**0.712**	**+13.2**	**+8.0**
es	0.568	0.600	**0.636**	**+6.8**	**+3.6**
zh	0.624	0.624	0.596	-2.8	-2.8
sw	0.320	0.332	0.296	-2.4	-3.6
bn	0.564	0.564	0.480	-8.4	-8.4
Mean	0.531	0.550	0.544	+1.3	-0.6

Δ values in absolute percentage points.

Mean accuracy (0.544) matches Zhao's subspace baseline (0.550), confirming feature-level ablation replicates the subspace effect. SAE outperforms Zhao on en (+8.0pp) and es (+3.6pp): per-feature targeting is more precise than subspace projection for some languages.

But the asymmetry is striking: en +13.2pp and es +6.8pp *gain*, while bn -8.4pp and sw -2.4pp *lose*. Why? Section 3B opens up the top-5 features per language and shows that different languages have qualitatively different feature taxonomies.

B. Which features affect performance?

Not all statistically-identified candidates behave alike. We ablate each of the top-5 A$\cap$B features per language individually and measure (i) perplexity delta on target-language text and (ii) accuracy delta on MGSM reasoning. A feature is tagged **LANGUAGE** if ablation raises perplexity but does not hurt accuracy (language-only); **SHARED** if it raises perplexity *and* drops accuracy (language and reasoning entangled).

Lang	LANG	SHARED	Other
en	**5**	0	0
es	**3**	0	2
zh	3	2	0
sw	2	**3**	0
bn	0	**5**	0

English's top-5 are all cleanly LANGUAGE; Bengali's are all SHARED. Languages whose top features cleanly separate from reasoning gain accuracy when ablated (en, es); those whose top features entangle with reasoning lose accuracy (bn, sw). The 3A asymmetry and the 3B taxonomy are **two views of the same phenomenon**; §3C now makes the mechanism explicit.

C. What happens to the reasoning features?

We ablate the same language features at L17 and measure how the 21 reasoning features respond (forward-only, 50 problems $\times$ 5 langs, Wilcoxon signed-rank), in the spirit of feature-circuit attribution [7]. Neuronpedia look-ups reveal three functional camps:

Camp	Neuronpedia identity	Effect
Reasoning (4094, 5993, 1105)	solving for x, step-by-step, causal reasoning	**Stable** $\lvert\Delta\rvert<200$
Answering (96, 406, 377, 712, 9240)	explanatory, QA patterns, response initiation	**Boosted** Δ up to +7036
Answering (107, 279, ...)	conversational acknowledgment, preamble ("Okay,...")	**Suppressed** to $\approx$0

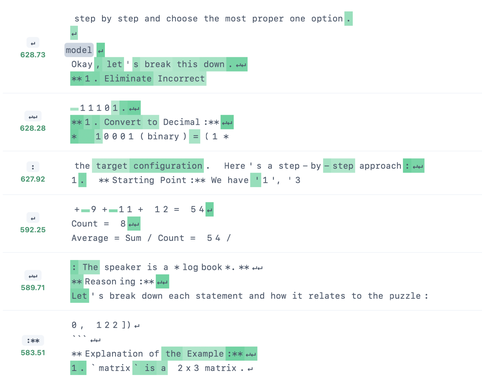

Figure 1. Feature 1105 (L17): fires on causal/sequential reasoning ("let's break this down"). **Unaffected** by language-feature ablation ($\Delta<200$).

Ablation reorganizes the answering pathway while reasoning features remain stable ($|\Delta|<200$ for 4094, 5993, 1105 across all 5 langs). Explanatory features get boosted while conversational preamble is suppressed, shifting the model toward direct answer generation. This explains the en/es asymmetry: clean LANGUAGE features ablate without side effects; SHARED features (bn/sw) disrupt reasoning along with language, blocking the gain.

4. But What Does It Mean to Be a Language Feature?

Two definitions of "language feature" are in play. **(i) Statistical:** a feature whose activation differs across languages (the output of our selection procedure). **(ii) Semantic:** a feature whose top activations are interpretable by a human as encoding a specific language. We evaluate both definitions along two axes: human recognizability via Neuronpedia auto-interp (§4A), and out-of-domain stability via FLORES (§4B). The two definitions return substantially different feature sets.

A. Humanly-interpretable features (via Neuronpedia)

We look up every chosen feature on Neuronpedia, an automated interpretability database that labels SAE features by top-K activating examples (the standard auto-interp protocol from [1]). This protocol characterizes peak activations and may miss features whose mass sits in the moderate-magnitude tail; the Sonnet study [8] flags this as an open limitation. We treat Neuronpedia labels as hypotheses for causal validation. Even for **English and Spanish**, our best-performing languages, only 2–3 of the top features are humanly interpretable as language-specific. For **Bengali and Swahili**, *none* map to recognizable language markers.

We hand-selected only Neuronpedia-confirmed features and ablated them (5–6 per language, n=50, both clamped and directional methods). **Result: no detectable change** ($\pm$0–4pp, indistinguishable from random ablation). The causally effective features are the statistically-identified ones lacking clear interpretability; Neuronpedia-validated picks alone produce null, confirming that the causal LANGUAGE/SHARED taxonomy carries the intervention effect.

B. Are these language features global?

Both Zhao's subspace and our SAE features are identified on math content. We run the *same* identification procedure on FLORES-200 (1012 parallel translations, no math) and compare.

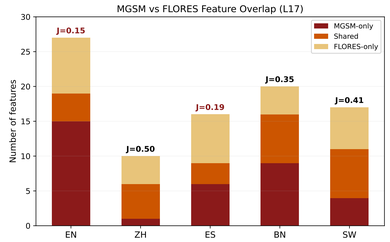

Figure 2. Same procedure, different domain $\rightarrow$ different features. English (J=0.15) and Spanish (J=0.19) share almost none.

The features identified as "language features" on MGSM are largely **different** from those identified on FLORES. A language probe trained on MGSM features transfers poorly to FLORES (accuracy 0.714, need $>$0.80), and the LANGUAGE/SHARED distinction from Section 3B is meaningless out of domain (p=0.51). If these were genuinely encoding *language identity* in the domain-general sense documented for multilingual LLMs [9], they would generalize across domains. The Jaccard overlap and probe-transfer results indicate that they encode **task-specific structure that co-varies with language** on a particular dataset.

5. Conclusion

1. **"Language feature" is an implicit label.** It is induced by the selection procedure (monolinguality $\cap$ probe); a different procedure (top-K activations on Neuronpedia) returns a different feature set. No operationalization is privileged.
2. **The procedure works.** Statistical identification + ablation reproduces and extends Zhao's gains (en +13.2pp, es +6.8pp).
3. **The features behave as MGSM-specific structure.** They fail to generalize out of domain (probe transfer 0.714, separation p=0.51), disagree with Neuronpedia auto-interp, and change identity across domains (en Jaccard=0.15). Their scope is bounded to the dataset on which they were identified.
4. **"Reasoning feature" is also a procedural tag.** The most dramatic released feature (96, Δ up to +7036) responds to narrative phrasing in word problems and activates only weakly on arithmetic operators, placing it in the answering camp. Both labels are method-dependent.

Discussion: when a feature fires vs. what it does

Correlational measures (top-activating examples, monolinguality, probe coefficients, auto-interp) characterize *when* a feature fires; characterizing *what* it does requires causal intervention. The label depends jointly on the selection procedure and the evaluative measure; we treat such labels as hypotheses for causal validation.

Limitations & future work

Single model (Gemma 3 4B IT), single task (MGSM), 5 languages; global k=20 suboptimal; dev/test overlap in profiling step. Next: compare across operationalizations (causal LM ablation, top-activating examples, activation steering); confirmed-only ablation using only causally-validated features per language.

References